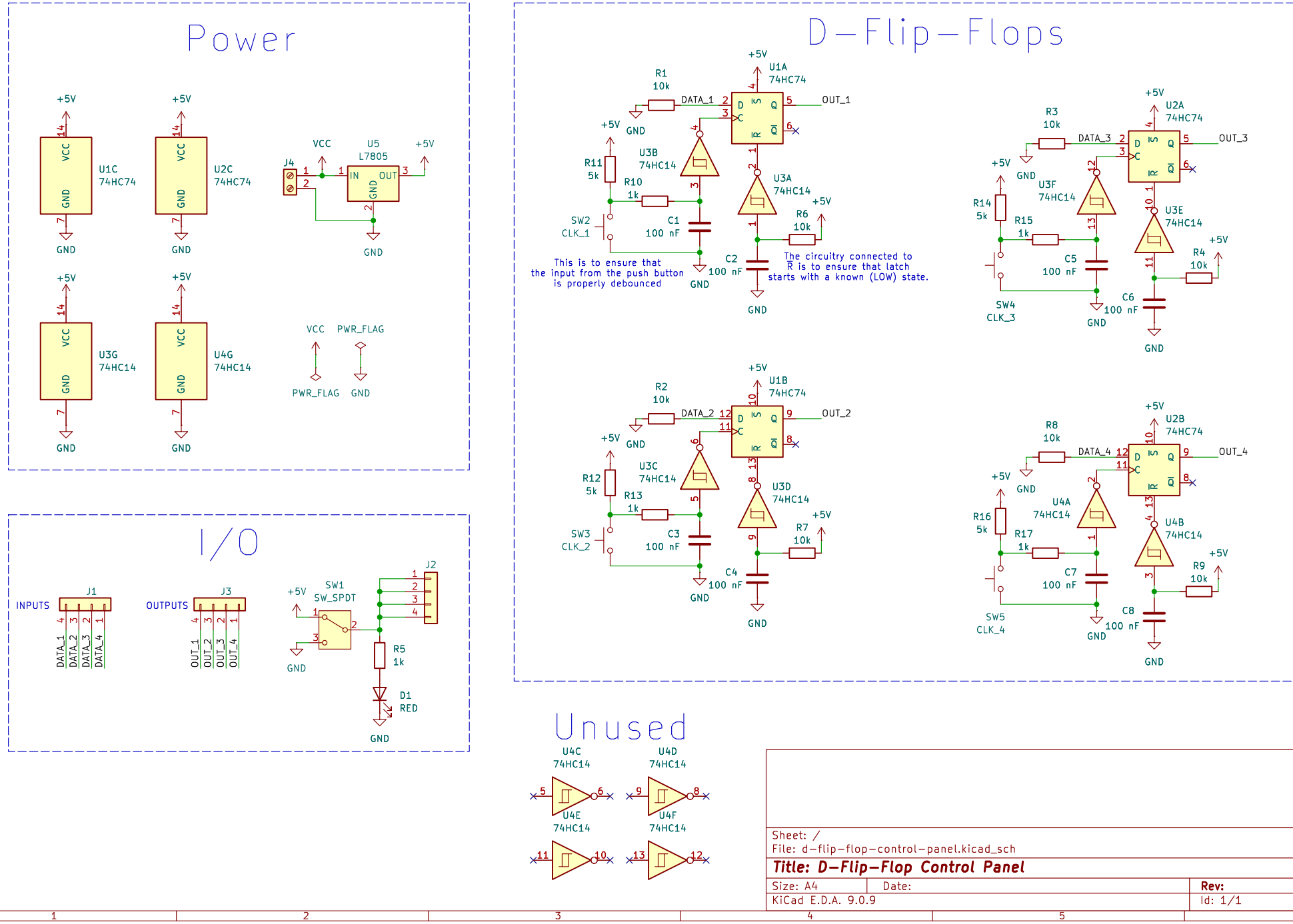

Power
+5V
U1C
74HC74
VCC
GND
U2C
74HC74
U5
L7805
IN
OUT
J4
U3G
74HC14
U4G
74HC14
PWR_FLAG
I/O
INPUTS
J1
DATA_1
DATA_2
DATA_3
DATA_4
OUTPUTS
J3
OUT_1
OUT_2
OUT_3
OUT_4
SW1
SW_SPDT
J2
R5
1k
D1
RED
D-Flip-Flops
R1
10k
U1A
74HC74
OUT_1
U3B
74HC14
U3A
74HC14
R11
5k
R10
1k
SW2
CLK_1
C1
100 nF
C2
100 nF
R6
10k
This is to ensure that the input from the push button is properly debounced
The circuitry connected to R is to ensure that latch starts with a known (LOW) state.
R2
10k
U1B
74HC74
OUT_2
U3C
74HC14
U3D
74HC14
R12
5k
R13
1k
SW3
CLK_2
C3
100 nF
C4
100 nF
R7
10k
R3
10k
U2A
74HC74
OUT_3
U3F
74HC14
U3E
74HC14
R14
5k
R15
1k
SW4
CLK_3
C5
100 nF
C6
100 nF
R4
10k
R8
10k
U2B
74HC74
OUT_4
U4A
74HC14
U4B
74HC14
R16
5k
R17
1k
SW5
CLK_4
C7
100 nF
C8
100 nF
R9
10k
Unused
U4C
74HC14
U4D
74HC14
U4E
74HC14
U4F
74HC14
Sheet: /
File: d-flip-flop-control-panel.kicad_sch
Title: D-Flip-Flop Control Panel
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.9
Id: 1/1